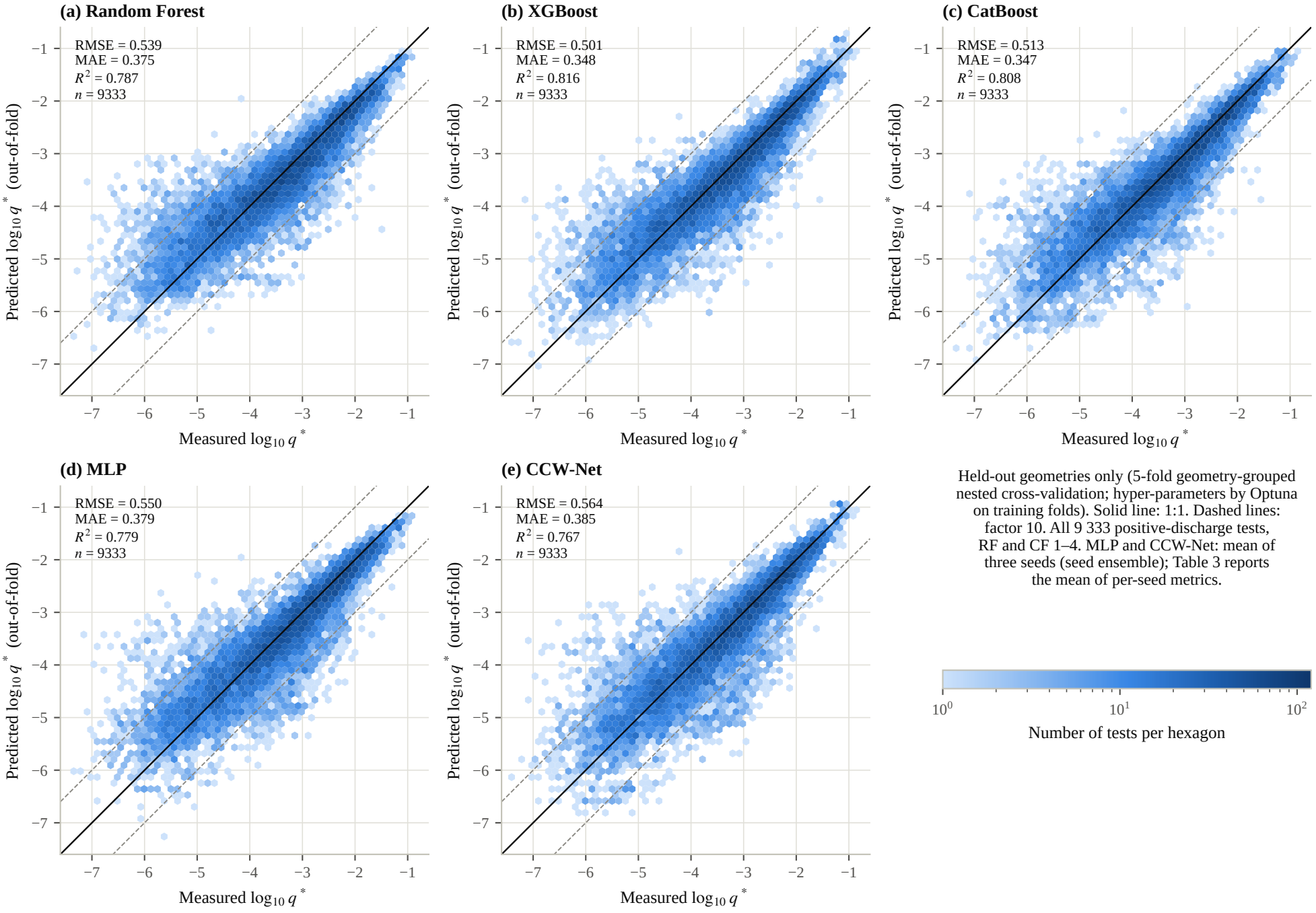

(a) Random Forest
RMSE = 0.539
MAE = 0.375
R^2 = 0.787
n = 9333
(b) XGBoost
RMSE = 0.501
MAE = 0.348
R^2 = 0.816
n = 9333
(c) CatBoost
RMSE = 0.513
MAE = 0.347
R^2 = 0.808
n = 9333
(d) MLP
RMSE = 0.550
MAE = 0.379
R^2 = 0.779
n = 9333
(e) CCW-Net
RMSE = 0.564
MAE = 0.385
R^2 = 0.767
n = 9333
Predicted $\log_{10} q^*$ (out-of-fold)
Measured $\log_{10} q^*$
−1
−2
−3
−4
−5
−6
−7
Held-out geometries only (5-fold geometry-grouped nested cross-validation; hyper-parameters by Optuna on training folds). Solid line: 1:1. Dashed lines: factor 10. All 9 333 positive-discharge tests, RF and CF 1–4. MLP and CCW-Net: mean of three seeds (seed ensemble); Table 3 reports the mean of per-seed metrics.
10^0
10^1
10^2
Number of tests per hexagon